1 Results from the zero-shot prompt.

Notation and interpretation of the risk. Let

$$\mathcal{R} := \mathcal{R}(t) \subseteq [p], \qquad q := |\mathcal{R}|, \qquad \mathcal{U} := \mathcal{R}^c,$$

and write the covariate as

$$x = (z^\top, u^\top)^\top, \qquad z := x_{\mathcal{R}} \in \mathbb{R}^q, \quad u := x_{\mathcal{U}} \in \mathbb{R}^{p-q}.$$

Partition the covariance matrix as

$$\Sigma = \begin{pmatrix} \Sigma_{RR} & \Sigma_{RU} \\ \Sigma_{UR} & \Sigma_{UU} \end{pmatrix}, \qquad \Sigma_{RR} \in \mathbb{R}^{q\times q}.$$

For the target site, write

$$\beta^{(t)} = \begin{pmatrix} \beta_R^{(t)} \\ \beta_U^{(t)} \end{pmatrix}, \qquad \beta^{(s)} = \begin{pmatrix} \beta_R^{(s)} \\ \beta_U^{(s)} \end{pmatrix}.$$

The estimator in the statement should use the available source estimator $\hat{\beta}^{(s)}$; accordingly we interpret

$$\hat{\omega}_s^{(t)} = \Phi^{(t)}\hat{\beta}^{(s)}, \qquad \Phi^{(t)} = \left[I_q,\ \Sigma_{RR}^{-1}\Sigma_{RU}\right].$$

This is the natural transfer center induced by the source estimate.

Since $\hat{\omega}_{\mathrm{TL}}^{(t)}$ depends on the target responses $Y^{(t)}$ and on $\hat{\beta}^{(s)}$, the quantity

$$r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) = \mathbb{E}\Big[\big(y^{(t)} - z^\top \hat{\omega}_{\mathrm{TL}}^{(t)}\big)^2 \,\Big|\, X_R^{(t)}\Big]$$

is still random through the training responses and the source estimator. The standard object in random-design asymptotics is therefore its conditional mean

$$\bar{r}^{\mathrm{pred}}(\lambda_\omega \mid X_R^{(t)}) := \mathbb{E}\Big[r^{\mathrm{pred}}\big(\hat{\omega}_{\mathrm{TL}}^{(t)}\big) \,\Big|\, X_R^{(t)}\Big],$$

where the outer expectation is over the remaining randomness ($Y^{(t)}$, $\beta^{(t)}$, $\beta^{(s)}$, $\hat{\beta}^{(s)}$). We shall derive a finite-sample exact decomposition for $\bar{r}^{\mathrm{pred}}$, then a high-dimensional asymptotic upper bound, and then the oracle choice of λ_ω.

Assumptions. We use the following regularity conditions.

(A1) **High-dimensional regime.** As $n_t, p \to \infty$,

$$\frac{p}{n_t} \to \gamma \in (0,\infty), \qquad \frac{q}{p} \to \kappa \in (0,1], \qquad \frac{q}{n_t} \to \eta := \kappa\gamma.$$

(A2) **Covariance regularity.** The matrix Σ is positive definite, and there exist constants $0 < c_\Sigma < C_\Sigma < \infty$ such that for all p,

$$c_\Sigma \le \lambda_{\min}(\Sigma) \le \lambda_{\max}(\Sigma) \le C_\Sigma.$$

Hence also

$$c_\Sigma \le \lambda_{\min}(\Sigma_{RR}) \le \lambda_{\max}(\Sigma_{RR}) \le C_\Sigma.$$

(A3) **Transfer map regularity.** There exists a finite constant C_Φ such that

$$\|\Phi^{(t)}\|_{\mathrm{op}}^2 = \lambda_{\max}\big(\Phi^{(t)}(\Phi^{(t)})^\top\big) \le C_\Phi \qquad \text{for all } p.$$

(A4) **Random coefficients.**

$$\mathbb{E}[\beta^{(m)}] = 0, \qquad \mathrm{Var}(\beta^{(m)}) = \frac{\alpha_m^2}{p} I_p, \qquad m \in \{s, t\},$$

and

$$\mathrm{Cov}\big(\beta^{(t)}, \beta^{(s)}\big) = \frac{\rho \alpha_t \alpha_s}{p} I_p, \qquad -1 \le \rho \le 1.$$

(A5) **Source estimation error.** Let

$$e_s := \hat{\beta}^{(s)} - \beta^{(s)}.$$

Assume e_s is independent of the target sample $(X^{(t)}, Y^{(t)})$ and of $\beta^{(t)}$, satisfies $\mathbb{E}[e_s] = 0$, and

$$\lambda_{\max}\Big(\mathbb{E}[e_s e_s^\top]\Big) \le \frac{C^U}{p}$$

for a constant $C^U > 0$.

(A6) **Noise.** The target noise variables ε_i are i.i.d., mean zero, variance σ_ε^2, and independent of x_i, $\beta^{(t)}$, $\beta^{(s)}$, and e_s.

(A7) **Spectral convergence of the observed design.** The empirical spectral distribution of

$$S_n := \frac{1}{n_t} (X_R^{(t)})^\top X_R^{(t)}$$

converges almost surely to a deterministic compactly supported probability law F_η. This is automatic, for example, if $z_i \sim N(0, \Sigma_{RR})$ and the empirical spectral distribution of Σ_{RR} converges, in which case F_η is the corresponding deformed Marchenko–Pastur limit.

Step 1: the target optimal linear predictor from $z = x_{\mathcal{R}}$.

Lemma 1 (Optimal observed-feature coefficient)**.** *Under the Gaussian model,*

$$\omega_*^{(t)} := \arg\min_{\omega \in \mathbb{R}^q} \mathbb{E}\Big[(y^{(t)} - z^\top \omega)^2 \,\Big|\, \beta^{(t)}\Big] = \Phi^{(t)} \beta^{(t)} = \beta_R^{(t)} + \Sigma_{RR}^{-1} \Sigma_{RU} \beta_U^{(t)}.$$

Moreover, writing

$$\Sigma_{U|R} := \Sigma_{UU} - \Sigma_{UR} \Sigma_{RR}^{-1} \Sigma_{RU},$$

we have the conditional representation

$$y^{(t)} = z^\top \omega_*^{(t)} + \xi,$$

where ξ is independent of z given $\beta^{(t)}$, with

$$\mathbb{E}[\xi \mid \beta^{(t)}] = 0, \qquad \mathrm{Var}(\xi \mid \beta^{(t)}) = \tau_t^2(\beta^{(t)}) := \sigma_\varepsilon^2 + (\beta_U^{(t)})^\top \Sigma_{U|R} \beta_U^{(t)}.$$

Proof. Since (z, u) is jointly Gaussian with mean 0,

$$\mathbb{E}[u \mid z] = \Sigma_{UR}\Sigma_{RR}^{-1} z.$$

Therefore

$$\mathbb{E}[y^{(t)} \mid z, \beta^{(t)}] = z^\top \beta_R^{(t)} + \mathbb{E}[u \mid z]^\top \beta_U^{(t)} = z^\top \left(\beta_R^{(t)} + \Sigma_{RR}^{-1}\Sigma_{RU}\beta_U^{(t)}\right).$$

The coefficient of the conditional mean is therefore $\omega_*^{(t)} = \Phi^{(t)}\beta^{(t)}$. The decomposition

$$\xi := y^{(t)} - z^\top \omega_*^{(t)} = u^\top \beta_U^{(t)} - \mathbb{E}[u^\top \beta_U^{(t)} \mid z] + \varepsilon$$

has conditional mean zero. Since, conditional on $\beta^{(t)}$, the Gaussian residual $u - \mathbb{E}[u \mid z]$ is independent of z and has covariance $\Sigma_{U|R}$,

$$\mathrm{Var}(\xi \mid \beta^{(t)}) = (\beta_U^{(t)})^\top \Sigma_{U|R}\beta_U^{(t)} + \sigma_\varepsilon^2.$$

This proves the claim. □

Step 2: closed form of the transfer-ridge estimator. Set

$$X := X_R^{(t)} \in \mathbb{R}^{n_t \times q}, \qquad Y := Y^{(t)} \in \mathbb{R}^{n_t}, \qquad S_n := \frac{1}{n_t} X^\top X.$$

The estimator is

$$\hat{\omega}_{\mathrm{TL}} = \arg\min_{\omega \in \mathbb{R}^q} \left[\frac{1}{n_t}\|Y - X\omega\|_2^2 + \lambda_\omega \|\omega - \hat{\omega}_s\|_2^2\right].$$

Its first-order condition gives

$$(S_n + \lambda_\omega I_q)\hat{\omega}_{\mathrm{TL}} = \frac{1}{n_t} X^\top Y + \lambda_\omega \hat{\omega}_s,$$

hence

$$\hat{\omega}_{\mathrm{TL}} = (S_n + \lambda_\omega I_q)^{-1} \left(\frac{1}{n_t} X^\top Y + \lambda_\omega \hat{\omega}_s\right).$$

Using $Y = X\omega_* + \xi$, where $\omega_* = \omega_*^{(t)}$ and $\xi = (\xi_1, \ldots, \xi_{n_t})^\top$, we obtain

$$\hat{\omega}_{\mathrm{TL}} - \omega_* = -\lambda_\omega A_n(\lambda_\omega)\delta + A_n(\lambda_\omega)\frac{1}{n_t} X^\top \xi,$$

where

$$A_n(\lambda) := (S_n + \lambda I_q)^{-1}, \qquad \delta := \omega_* - \hat{\omega}_s.$$

This is the fundamental bias–variance decomposition.

Step 3: exact finite-sample conditional mean predictive risk.

Theorem 1 (Exact decomposition)**.** *Under (A1)–(A6), for every fixed $\lambda > 0$,*

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) = \bar{\tau}_t^2 + \lambda^2 \operatorname{tr}\Big\{\Sigma_{RR} A_n(\lambda)\, \mathbb{E}[\delta\delta^\top]\, A_n(\lambda)\Big\} + \frac{\bar{\tau}_t^2}{n_t} \operatorname{tr}\{\Sigma_{RR} A_n(\lambda) S_n A_n(\lambda)\}\,,$$

where

$$\bar{\tau}_t^2 := \mathbb{E}[\tau_t^2(\beta^{(t)})] = \sigma_\varepsilon^2 + \frac{\alpha_t^2}{p}\operatorname{tr}(\Sigma_{U|R}).$$

Proof. For a fresh test point $(z_{\mathrm{new}}, y_{\mathrm{new}})$, conditional on $\beta^{(t)}$,

$$y_{\mathrm{new}} = z_{\mathrm{new}}^\top \omega_* + \xi_{\mathrm{new}},$$

with ξ_{new} independent of z_{new} and of the training sample, and $\mathbb{E}[\xi_{\mathrm{new}}^2 \mid \beta^{(t)}] = \tau_t^2(\beta^{(t)})$. Hence, conditional on X,

$$r^{\mathrm{pred}}(\hat{\omega}_{\mathrm{TL}}) = \tau_t^2(\beta^{(t)}) + (\hat{\omega}_{\mathrm{TL}} - \omega_*)^\top \Sigma_{RR} (\hat{\omega}_{\mathrm{TL}} - \omega_*).$$

Taking the outer conditional expectation given X yields

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) = \bar{\tau}_t^2 + \mathbb{E}\Big[(\hat{\omega}_{\mathrm{TL}} - \omega_*)^\top \Sigma_{RR} (\hat{\omega}_{\mathrm{TL}} - \omega_*) \,\Big|\, X\Big]\,.$$

Insert the identity

$$\hat{\omega}_{\mathrm{TL}} - \omega_* = -\lambda A_n(\lambda)\delta + A_n(\lambda)\frac{1}{n_t} X^\top \xi.$$

Because $\mathbb{E}[\delta] = 0$ and δ is independent of ξ conditional on X, the cross term has mean zero. Therefore

$$\mathbb{E}\Big[(\hat{\omega}_{\mathrm{TL}} - \omega_*)^\top \Sigma_{RR} (\hat{\omega}_{\mathrm{TL}} - \omega_*) \,\Big|\, X\Big] = \lambda^2 \operatorname{tr}\Big\{\Sigma_{RR} A_n(\lambda) \mathbb{E}[\delta\delta^\top] A_n(\lambda)\Big\} + \mathbb{E}\Big[\Big(\frac{1}{n_t} X^\top \xi\Big)^\top A_n(\lambda) \Sigma_{RR} A_n(\lambda) \Big(\frac{1}{n_t} X^\top \xi\Big) \,\Big|\, X$$

Now conditional on X,

$$\operatorname{Var}\Big(\frac{1}{n_t} X^\top \xi \,\Big|\, X\Big) = \frac{1}{n_t^2} X^\top \mathbb{E}[\xi\xi^\top] X.$$

Since the ξ_i are independent, mean zero, and each has variance $\bar{\tau}_t^2$ after averaging over $\beta^{(t)}$,

$$\mathbb{E}[\xi\xi^\top] = \bar{\tau}_t^2 I_{n_t}.$$

Hence

$$\operatorname{Var}\Big(\frac{1}{n_t} X^\top \xi \,\Big|\, X\Big) = \frac{\bar{\tau}_t^2}{n_t} S_n.$$

Using $\mathbb{E}[v^\top M v] = \operatorname{tr}(M \operatorname{Var}(v))$ for zero-mean v,

$$\mathbb{E}\Big[\Big(\frac{1}{n_t} X^\top \xi\Big)^\top A_n(\lambda) \Sigma_{RR} A_n(\lambda) \Big(\frac{1}{n_t} X^\top \xi\Big) \,\Big|\, X\Big] = \frac{\bar{\tau}_t^2}{n_t} \operatorname{tr}\{\Sigma_{RR} A_n(\lambda) S_n A_n(\lambda)\}\,.$$

This proves the formula. □

Step 4: control of the transfer mismatch term $\mathbb{E}[\delta\delta^\top]$.

Lemma 2 (Second-moment bound for the transfer mismatch)**.** *Define*

$$\Delta_{ts}^2 := \alpha_t^2 + \alpha_s^2 - 2\rho\alpha_t\alpha_s.$$

Under (A3)–(A5),

$$\mathbb{E}[\delta\delta^\top] \preceq \frac{\Delta_{ts}^2 + C^U}{p}\,\Phi^{(t)}(\Phi^{(t)})^\top \preceq \frac{(\Delta_{ts}^2 + C^U)C_\Phi}{p}\,I_q.$$

Proof. By definition,

$$\delta = \omega_* - \hat{\omega}_s = \Phi^{(t)}\beta^{(t)} - \Phi^{(t)}\hat{\beta}^{(s)} = \Phi^{(t)}\Big[(\beta^{(t)} - \beta^{(s)}) - e_s\Big].$$

Since e_s is mean zero and independent of $(\beta^{(t)}, \beta^{(s)})$,

$$\mathbb{E}[\delta\delta^\top] = \Phi^{(t)}\left\{\mathrm{Var}(\beta^{(t)} - \beta^{(s)}) + \mathbb{E}[e_s e_s^\top]\right\}(\Phi^{(t)})^\top.$$

Now

$$\mathrm{Var}(\beta^{(t)} - \beta^{(s)}) = \frac{\alpha_t^2}{p}I_p + \frac{\alpha_s^2}{p}I_p - 2\frac{\rho\alpha_t\alpha_s}{p}I_p = \frac{\Delta_{ts}^2}{p}I_p.$$

By (A5),

$$\mathbb{E}[e_s e_s^\top] \preceq \frac{C^U}{p}I_p.$$

Therefore

$$\mathbb{E}[\delta\delta^\top] \preceq \frac{\Delta_{ts}^2 + C^U}{p}\,\Phi^{(t)}(\Phi^{(t)})^\top.$$

The second inequality follows from $\|\Phi^{(t)}\|_{\mathrm{op}}^2 \le C_\Phi$. □

Step 5: finite-sample upper bound.

Theorem 2 (Finite-sample conditional upper bound)**.** *Under (A1)–(A6), for every $\lambda > 0$,*

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + \frac{(\Delta_{ts}^2 + C^U)C_\Phi}{p}\,\lambda^2\,\mathrm{tr}\big\{\Sigma_{RR}A_n(\lambda)^2\big\} + \frac{\bar{\tau}_t^2}{n_t}\,\mathrm{tr}\big\{\Sigma_{RR}A_n(\lambda)S_nA_n(\lambda)\big\}.$$

In particular, by $\lambda_{\max}(\Sigma_{RR}) \le C_\Sigma$,

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\frac{\lambda^2}{p}\,\mathrm{tr}\big\{A_n(\lambda)^2\big\} + C_\Sigma\bar{\tau}_t^2\frac{1}{n_t}\,\mathrm{tr}\{A_n(\lambda)S_nA_n(\lambda)\big\}.$$

Proof. Start from the exact decomposition in the previous theorem and use the matrix bound from the lemma:

$$\lambda^2\mathrm{tr}\Big\{\Sigma_{RR}A_n(\lambda)\mathbb{E}[\delta\delta^\top]A_n(\lambda)\Big\} \le \frac{(\Delta_{ts}^2 + C^U)C_\Phi}{p}\,\lambda^2\,\mathrm{tr}\big\{\Sigma_{RR}A_n(\lambda)^2\big\}.$$

This gives the first inequality. The second follows from

$$\Sigma_{RR} \preceq C_\Sigma I_q.$$

□

Step 6: asymptotic deterministic equivalent. Define for $\lambda > 0$,

$$\frac{1}{q}\mathrm{tr}\{A_n(\lambda)^2\} \xrightarrow{\text{a.s.}} v_\eta(\lambda), \qquad \frac{1}{q}\mathrm{tr}\{A_n(\lambda)S_nA_n(\lambda)\} \xrightarrow{\text{a.s.}} u_\eta(\lambda).$$

Using $q/p \to \kappa$ and $q/n_t \to \eta$, the finite-sample upper bound immediately yields:

Theorem 3 (High-dimensional asymptotic upper bound)**.** *Under (A1)–(A7), for every fixed $\lambda > 0$,*

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + \kappa\, C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\, \lambda^2 v_\eta(\lambda) + \eta\, C_\Sigma \bar{\tau}_t^2\, u_\eta(\lambda) + o_{\mathbb{P}}(1).$$

Equivalently,

$$\limsup_{n_t,p\to\infty} \bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + \kappa\, C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\, \lambda^2 v_\eta(\lambda) + \eta\, C_\Sigma \bar{\tau}_t^2\, u_\eta(\lambda)$$

in probability.

Proof. From the preceding theorem,

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\frac{\lambda^2}{p}\,\mathrm{tr}\{A_n(\lambda)^2\} + C_\Sigma\bar{\tau}_t^2\frac{1}{n_t}\,\mathrm{tr}\{A_n(\lambda)S_nA_n(\lambda)\}.$$

Now write

$$\frac{1}{p}\mathrm{tr}\{A_n(\lambda)^2\} = \frac{q}{p}\cdot\frac{1}{q}\mathrm{tr}\{A_n(\lambda)^2\}, \qquad \frac{1}{n_t}\mathrm{tr}\{A_n(\lambda)S_nA_n(\lambda)\} = \frac{q}{n_t}\cdot\frac{1}{q}\mathrm{tr}\{A_n(\lambda)S_nA_n(\lambda)\}.$$

By (A1) and (A7),

$$\frac{1}{p}\mathrm{tr}\{A_n(\lambda)^2\} \to \kappa v_\eta(\lambda), \qquad \frac{1}{n_t}\mathrm{tr}\{A_n(\lambda)S_nA_n(\lambda)\} \to \eta u_\eta(\lambda),$$

in probability (indeed almost surely on the event of spectral convergence). Substituting these limits proves the claim. □

Oracle choice of λ_ω. The asymptotic upper bound is minimized by

$$\lambda_\omega^{\mathrm{opt}} \in \arg\min_{\lambda>0}\Big\{\kappa\, C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\lambda^2 v_\eta(\lambda) + \eta\, C_\Sigma\bar{\tau}_t^2\, u_\eta(\lambda)\Big\}.$$

Since $u_\eta(\lambda)$ is strictly decreasing in λ and $\lambda^2 v_\eta(\lambda)$ is increasing for small λ and bounded as $\lambda \to \infty$, the objective has the standard bias–variance tradeoff structure and admits at least one minimizer on $(0, \infty)$.

Differentiating,

$$\frac{d}{d\lambda}\Big[\lambda^2 v_\eta(\lambda)\Big] = 2\lambda v_\eta(\lambda) + \lambda^2 v_\eta'(\lambda),$$

and

$$u_\eta'(\lambda) = -2\int \frac{x}{(x+\lambda)^3}\, dF_\eta(x) < 0.$$

Hence any interior optimizer satisfies the first-order condition

$$\kappa\, C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\Big[2\lambda v_\eta(\lambda) + \lambda^2 v_\eta'(\lambda)\Big] + \eta\, C_\Sigma\bar{\tau}_t^2\, u_\eta'(\lambda) = 0.$$

Equivalently,

$$\kappa\, C_\Phi(\Delta_{ts}^2 + C^U)\Big[2\lambda v_\eta(\lambda) + \lambda^2 v_\eta'(\lambda)\Big] + \eta\,\bar{\tau}_t^2\, u_\eta'(\lambda) = 0.$$

This equation characterizes the asymptotically optimal regularization level.

Transparent special case: $\Sigma = I_p$. When $\Sigma = I_p$, the situation becomes especially simple:

$$\Sigma_{RR} = I_q, \qquad \Sigma_{RU} = 0, \qquad \Phi^{(t)} = [I_q, 0].$$

Thus

$$\omega_*^{(t)} = \beta_R^{(t)}, \qquad \hat{\omega}_s^{(t)} = \hat{\beta}_R^{(s)}, \qquad \Sigma_{U|R} = I_{p-q},$$

and therefore

$$\bar{\tau}_t^2 = \sigma_\varepsilon^2 + \alpha_t^2 \frac{p-q}{p} \;\longrightarrow\; \sigma_\varepsilon^2 + \alpha_t^2(1-\kappa).$$

Moreover $C_\Sigma = C_\Phi = 1$, and the asymptotic upper bound reduces to

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + \kappa(\Delta_{ts}^2 + C^U)\lambda^2 v_\eta^{\mathrm{MP}}(\lambda) + \eta\bar{\tau}_t^2\, u_\eta^{\mathrm{MP}}(\lambda) + o_{\mathbb{P}}(1),$$

where $u_\eta^{\mathrm{MP}}, v_\eta^{\mathrm{MP}}$ are computed under the Marchenko–Pastur law with aspect ratio η. Thus the oracle tuning parameter is

$$\lambda_{\omega,\mathrm{opt}} \in \arg\min_{\lambda>0}\Big\{\kappa(\Delta_{ts}^2 + C^U)\lambda^2 v_\eta^{\mathrm{MP}}(\lambda) + \eta\bar{\tau}_t^2 u_\eta^{\mathrm{MP}}(\lambda)\Big\}.$$

Interpretation of the bound. The bound has three terms:

$$\underbrace{\bar{\tau}_t^2}_{\text{irreducible error}} + \underbrace{\kappa\, C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\lambda^2 v_\eta(\lambda)}_{\text{transfer bias / mismatch}} + \underbrace{\eta\, C_\Sigma \bar{\tau}_t^2 u_\eta(\lambda)}_{\text{estimation variance}}.$$

The quantity

$$\Delta_{ts}^2 = \alpha_t^2 + \alpha_s^2 - 2\rho\alpha_t\alpha_s$$

is the intrinsic target–source discrepancy. It is small when the source and target coefficients are strongly aligned (ρ large), and large when the two sites are poorly matched. The additional term C^U quantifies source-estimation uncertainty. Thus transfer is beneficial precisely when

$$\Delta_{ts}^2 + C^U$$

is small enough that shrinking toward the source center decreases variance more than it introduces transfer bias.

Conclusion. Under the stated assumptions, the transfer-ridge estimator around the source-induced center $\hat{\omega}_s^{(t)} = \Phi^{(t)}\hat{\beta}^{(s)}$ admits the exact conditional mean predictive-risk decomposition

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) = \bar{\tau}_t^2 + \lambda^2 \operatorname{tr}\Big\{\Sigma_{RR} A_n(\lambda)\, \mathbb{E}[\delta\delta^\top]\, A_n(\lambda)\Big\} + \frac{\bar{\tau}_t^2}{n_t} \operatorname{tr}\{\Sigma_{RR} A_n(\lambda) S_n A_n(\lambda)\},$$

and hence the asymptotic upper bound

$$\bar{r}^{\mathrm{pred}}(\lambda \mid X) \le \bar{\tau}_t^2 + \kappa\, C_\Sigma C_\Phi(\Delta_{ts}^2 + C^U)\lambda^2 v_\eta(\lambda) + \eta\, C_\Sigma \bar{\tau}_t^2 u_\eta(\lambda) + o_{\mathbb{P}}(1).$$

The asymptotically optimal regularization level is the minimizer of the deterministic equivalent on the right-hand side, equivalently the solution of the displayed first-order equation.